

Aim: VLAN - Inter V-LAN

Problem Statement: Design and configure a network that uses multiple VLAN's to segment traffic. Implement inter-VLAN routing using a router-on-a-stick on layer-3 switch so that hosts in different VLAN's can communicate securely. Verify connectivity and ensure proper isolation between broadcast domains.

Theory: VLAN (Virtual Local Area Network)

A VLAN is like creating separate virtual rooms inside one physical network. Each department gets its own VLAN so;

- Their traffic stays separate.
- Broadcast traffic is reduced.
- Security improves.

Ex. VLAN 10 - Accounts

VLAN 20 - HR

VLAN 30 - IT

Even though they use the same switches and cables, they act like separate networks.

Problem: VLANs cannot talk to each other by default
Devices in different VLANs cannot communicate.

Router-on-a-stick - A router is connected to the switch with one trunk port. That trunk carries traffic from all VLANs and the router routes between them.

Layer-3 switch - A layer-3 switch.

A layer-3 switch can act like both:

- a switch (handles VLANs)
- a router (routes between VLANs)

This method is faster and often used in larger networks.

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Security and Isolation: Even though VLANs can communicate:

- Broadcasts are still isolated
- Access control lists (ACLs) can restrict who talks to who.

ex: HR can access server VLAN, but not IT VLAN.

Network Idea - (Router - on - a - stick)

PC 1 (VLAN 10)

(Account)

PC 2 (VLAN 20)

PC 3 (VLAN 30)

- Switches create VLANs (separate groups)
- Router connects them so they can talk safely)

Step - by - step - (Concept - what you do)

1. Create VLANs on the switch - Give each department its own VLAN.

VLAN 10 - Accounts

VLAN 20 - HR

VLAN 30 - IT

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2. Put each PC into the correct VLAN - Each switch port belongs to one VLAN.

3. Make one link to the router a trunk.
The trunk carries traffic for all VLANs.

4. Set inter VLAN routing on the router.

ex: configuration (cisco - very basic)

switch

vlan 10

name accounts

vlan 20

name HR

vlan 30

name IT

interface fa 0/1

switchport mode access

switchport access vlan 10

interface fa 0/2

switchport mode access

switchport access vlan 20

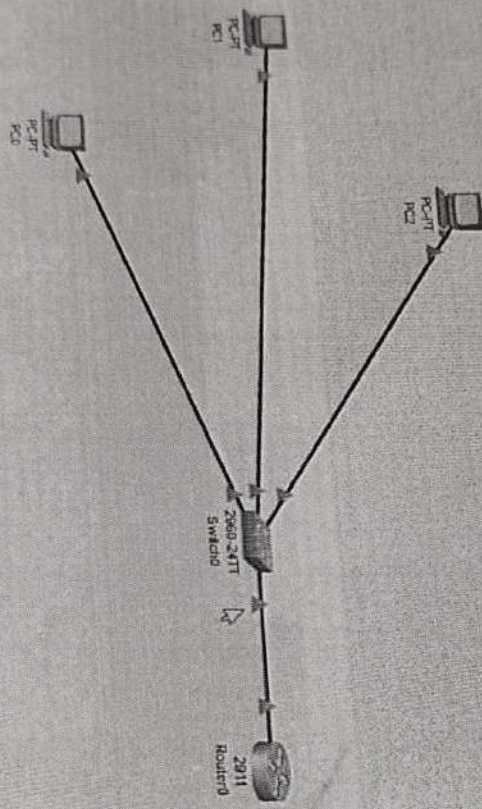
interface fa 0/3

switchport mode access

switchport access vlan 30

interface fa 0/24

switchport mode trunk



Router (Router - on - a - stick)

interface g0/0.10

encapsulation dot1q 10

ip address 192.168.10.1

255.255.255.0

interface g0/0.20

encapsulation dot1q 20

ip address 192.168.20.1

255.255.255.0

interface g0/0.30

encapsulation dot1q 30

ip address 192.168.30.1

255.255.255.0

Security - Use ACLs to control communication.

Allow HR → permit VLAN, block HR → IT

Conclusion: Hence, we shielded the VLAN -
inter VLAN.

~~08/11/20~~

Aim: Configuring Standard and extended ACLs.

problem statement: create and apply both standard and extended Access Control Lists (ACLs) to control traffic flow within a network. Implement rules to permit or deny specific hosts, subnets, protocols, and ports. Test the ACL functionality to ensure correct filtering without disrupting authorized communication.

Theory: What is an ACL (Access Control List)?

An ACL is a set of rules used in networking to allow or block traffic.

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Why do we use ACLs?

ACLs are used to:

- Improve network security
- Control who can access what
- Block unauthorized users.
- Allow only needed traffic.

Types of ACLs.

(1) Standard ACL: Standard ACL checks only the source IP address, who is sending the traffic.

It does not check:

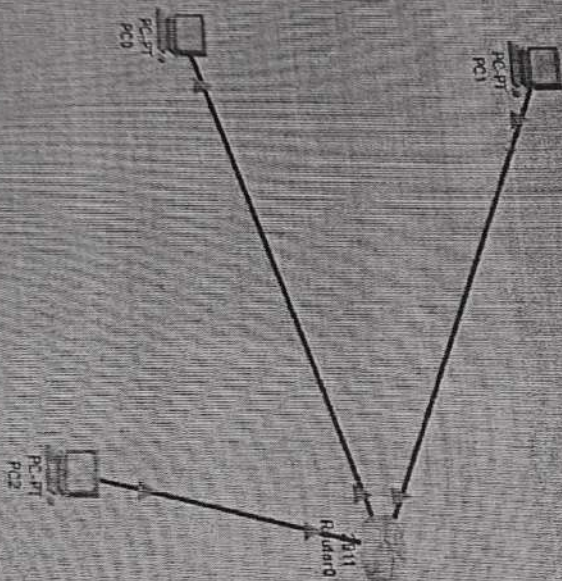
- Destination
- Protocol
- Port number.

ex: • Allow traffic from 192.168.1.10
• Block traffic from 192.168.1.20

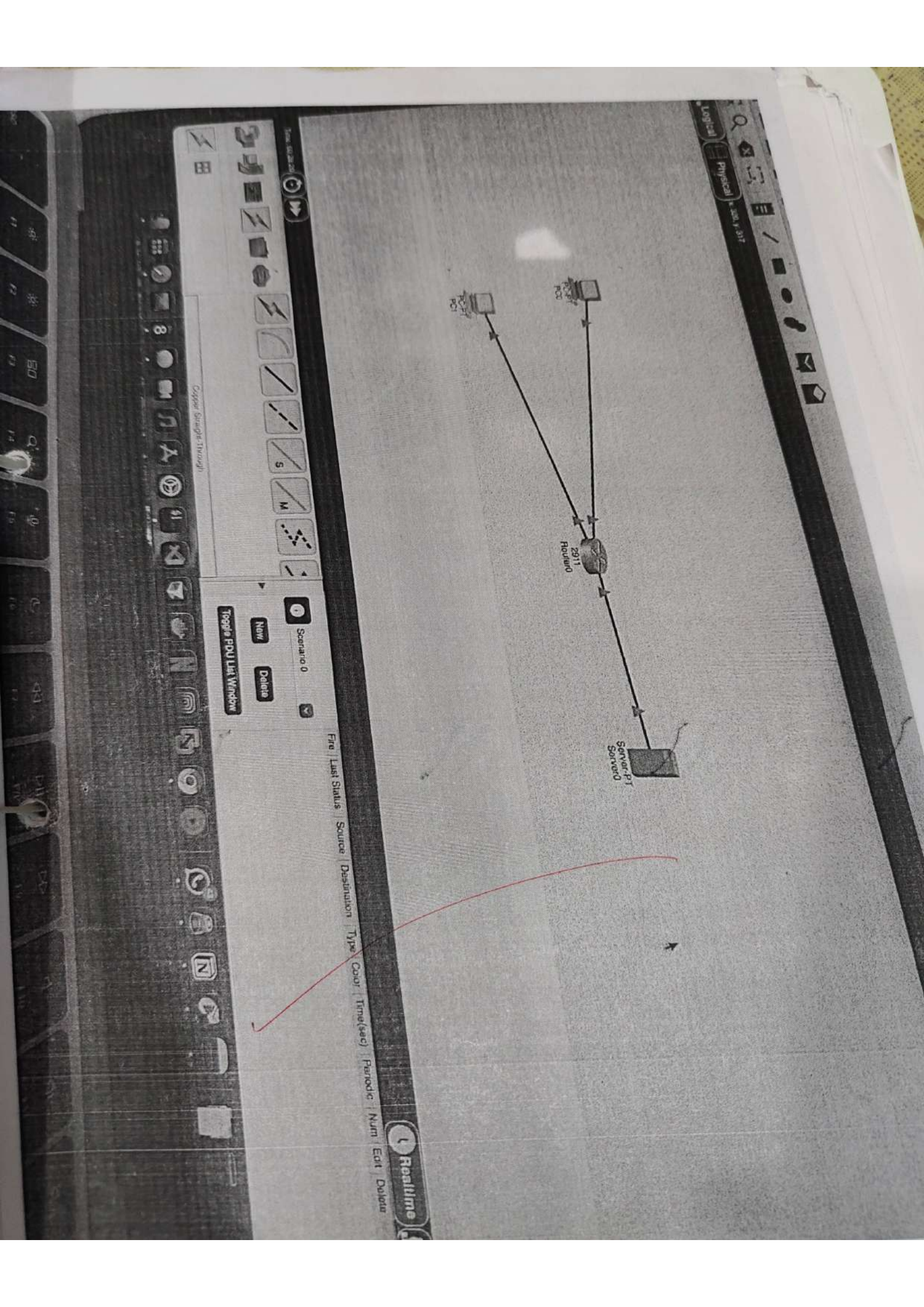
2] Extended ACL: extended ACL is more powerful.

- Source IP
- Destination IP
- Protocol (TCP, UDP, ICMP)
- Port number (HTTP, FTP, SSH etc)

ex: • Allow PC1 to access web server (port 80)
• Block PC2 from using FTP (port 21)



L2



What can ACL rules do?

ACL rules can : (1) permit/callow traffic.
(2) deny/block traffic.

How ACLs control traffic.

1. A packet enters the router.
2. ACL checks the packet
3. If rule matches - permit or deny.
4. If no rule matches - deny by default.

Testing ACL functionality.

After applying ACLs, we test to make sure

- Allowed users can communicate.
- Blocked users cannot access.
- No important service is disturbed.

Common testing methods:

- ping (ICMP)
- web access (HTTP/HTTPS)
- FTP or SSH test.
- Show ACL commands on router.

Conclusion: Hence, we studied, configuring standard and extended ACLs.

Aim: Routing protocols - RIP, EIGRP, OSPF
problem statement: Set up a multi-router environment and configure dynamic routing protocols (RIP, EIGRP, and OSPF). Compare their behavior in terms of convergence, metrics, and route selection. Verify that all routers have consistent routing tables and that end-to-end connectivity is achieved.

Theory: Routing is the process of finding the best path for data to travel from one network to another through routers. Routers use routing tables to decide where to send data packets.

What is Routing protocol - A routing protocol is a set of rules that use to share network information with other routers. Automatically learn routes. select the best path to reach a destination. Dynamic routing protocols update routes automatically when network changes occur.

Types of Routing protocols -
 1. Distance Vector - RIP
 2. Advanced Distance Vector (hybrid) - EIGRP
 3. Link State - OSPF

RIP (Routing Information Protocol) - RIP is a distance vector routing protocol that uses hop count as its metric. Maximum hop count: 15 (16 means unreachable). Sends routing updates every 30 seconds. Simple and easy to configure.

Metric Used: Hop Count (numbers of routers crossed)

Advantages:

1. Very easy to understand
2. Suitable for small networks.

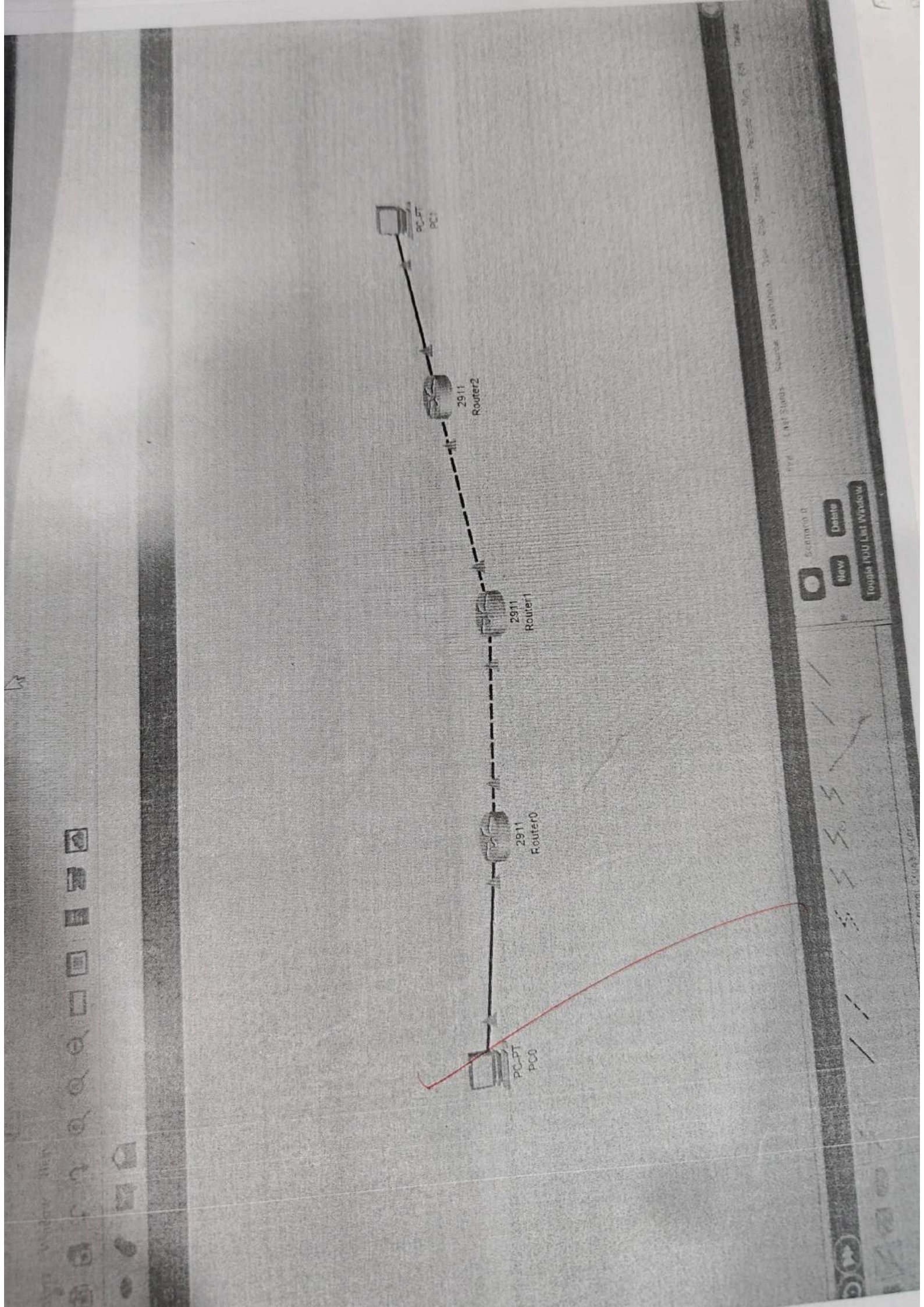
Disadvantages:

1. Not suitable for large networks
2. Slow convergence
3. Limited hop count.

EIGRP (Enhanced Interior Gateway Routing protocol) - EIGRP is an advanced distance vector (hybrid) routing protocol developed by Cisco.

Key points: Uses DUAL algorithm to find best path. Faster convergence than RIP. Supports large networks. Uses multiple metrics.

Metric Used: Bandwidth, Delay, Reliability, Load (mostly bandwidth & delay).



Item	Serial	Source	Destination	Type	Bandwidth	Delay	Cost	Table
1	0/0/0	192.168.1.1	192.168.1.2	Serial	1000	1000	1	Table
2	0/0/1	192.168.1.2	192.168.1.3	Serial	1000	1000	1	Table
3	0/0/2	192.168.1.3	192.168.1.4	Serial	1000	1000	1	Table

Scenario 1

☐ New ☐ Delete

Toggle Full List Window

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTRL/Z.

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router#
&SYS-5-CONFIG_I: Configured from console by console

Router#exit
```

router>show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
  10.0.0.0/30 is directly connected, GigabitEthernet0/1
  10.0.0.1/32 is directly connected, GigabitEthernet0/1
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
  192.168.1.0/24 is directly connected, GigabitEthernet0/0
  192.168.1.1/32 is directly connected, GigabitEthernet0/0
```

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#interface g0/0
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router#
```

&SYS-5-CONFIG_I: Configured from console by console

```
Router#exit
```

I

Router>show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
 * - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
 P - candidate default, U - per-user static route, o - ODR
 P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
 10.0.0.0/30 is directly connected, GigabitEthernet0/1
 10.0.0.2/32 is directly connected, GigabitEthernet0/1
 10.0.0.4/30 is directly connected, GigabitEthernet0/0
 10.0.0.5/32 is directly connected, GigabitEthernet0/0

Router>

Advantages: very fast convergence, efficient use of bandwidth, supports unequal cost balancing.

Disadvantages: Cisco proprietary (most Cisco devices)

OSPF (Open Shortest Path First)

OSPF is a link-state routing protocol that uses cost as its metric.

Key points: uses Dijkstra's shortest path algorithm.
Sends updates only when changes occur.
Divides network into areas.
very fast convergence.

Metric used: Cost (based on bandwidth)

Advantages: Scalable and reliable, fast convergence, suitable for large networks.

Disadvantages: more complex configurations.
Requires more memory & CPU.

Conclusion: Hence, we studied the Routing protocols - RIP, EIGRP, OSPF.

Aim: IPsec VPN with Advanced Configuration.

problem statement: Build a secure site-to-site IPsec VPN tunnel between two remote networks. Implement advanced features such as transform sets, ISAKMP policies, pre-shared keys and tunnel mode options. Ensure encrypted traffic passes between LANs and validate tunnel establishment through debugging commands.

Theory: A VPN (Virtual private network) creates a secure, encrypted connection over the Internet. It allows two private networks at different locations to communicate as if they are directly connected.

ARISE & SHINE

What is IPsec → IPsec (Internet protocol security) is a set of rules and protocols used to secure IP communication.

IPsec provides:

- (1) Encryption - protects data from hackers.
- (2) Authentication - verifies sender identity.
- (3) Integrity - ensures data is not changed.

23) Site-to-site IPsec VPN - A site-to-site VPN connects two entire LAN networks through the internet.

ex: (1) Office A LAN $\leftrightarrow$ Office B LAN

(2) All devices can communicate securely without installing VPN software on each PC.

(4) How IPsec VPN works :-

(1) Routers at both sites identify interesting traffic.

(2) ISAKMP negotiates security parameters.

(3) Authentication is done using pre-shared key.

(4) IPsec tunnel is created.

(5) Encrypted data starts flowing between LANs.

(5) ISAKMP (Internet Security Association and Key Management Protocol).

ISAKMP handles phase 1 of VPN.

It decides:

(1) Encryption algorithm.

(2) Hashing method.

(3) Authentication type.

(4) Diffie-Hellman group

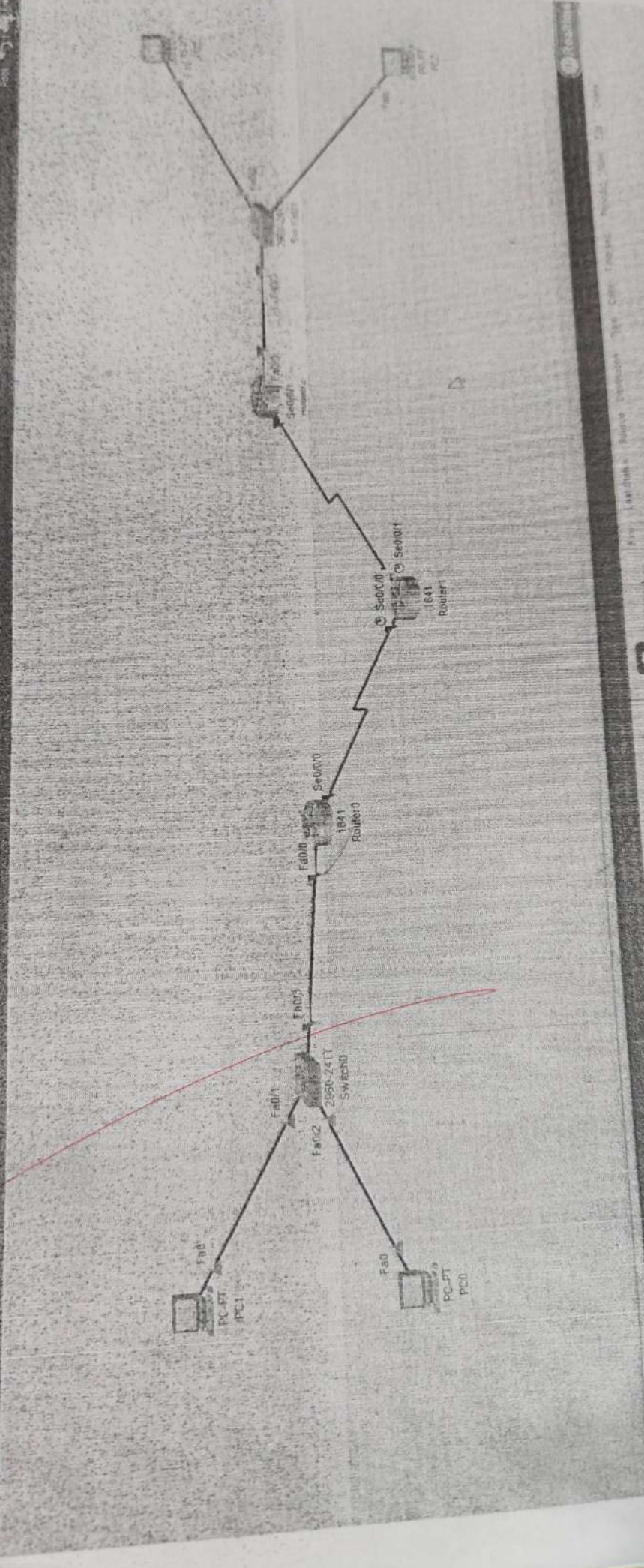
(5) Tunnel lifetime

Purpose: Build a secure tunnel channel.

Cisco Packet Tracer

File Edit View Tools Windows Help

Logical Physical



Time: 01:00:17

View: Topology, CLI, Config, Debug, Packet Tracer, Simulation, Erase, Reload, Shutdown, Start, Stop, Pause, Resume, Reset, Save, Load, Print, Help

Config

CLI

Attributes

IOS Command Line Interface

```
router>enable
router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
router(config)#crypto
Incomplete command.
router(config)#crypto isakmp
Incomplete command.
router(config)#crypto isakmp policy 10
router(config-isakmp)#encryption
Incomplete command.
router(config-isakmp)#encryption aes 256
router(config-isakmp)#hash sha256

Invalid input detected at '^' marker.
```

```
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#exit
Router(config)#crypto isakmp key?
key
Router(config)#crypto isakmp key
% Incomplete command.
Router(config)#crypto isakmp key?
key
Router(config)#crypto isakmp key?
key
Router(config)#crypto isakmp key?
key
Router(config)#crypto isakmp key
% Incomplete command.
Router(config)#crypto
% Invalid input detected at '^' marker.
```

```
Router(config)#crypto isakmp key cisco address 203.0.113.25
Router(config)#
```

IOS Command Line Interface

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CTRL-Z
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
%LINK-3-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#router rip
Router(config-router)#network 20.0.0.0
Router(config-router)#192.168.2.1 255.255.255.0
% Invalid input detected at ' ' marker
Router(config-router)#

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#router rip
Router(config-router)#network 10.0.0.1
Router(config-router)#network 20.0.0.2
Router(config-router)#

(6) ISAKMP policy - An ISAKMP policy defines how both routers agree on security settings.

Common Parameters - (1) Encryption: AES/3DES

(2) Hashing: SHA/MD5

(3) Authentication: Pre-shared key

(4) DH Group: Group 2, 5 etc.

(5) Lifetime: Time before re-negotiation.

Both routers must match these values.

(7) Pre-shared key (PSK) - A pre-shared key is a secret password shared between both routers. (1) Used for authentication.

(2) Must be same on both sides.

(3) ~~Means~~ only trusted devices connect.

(8) IPsec Transform set - A transform set defines how data is protected during transmission. (1) Encryption algorithm (ESP-AES, ESP-3DES)

(2) Authentication algorithm (SHA, MD5)

Used in Phase 2 of IPsec.

Conclusion: Hence we studied, IPsec VPN with Advanced Configuration.

~~24/2/26~~

practical No-05

Aim: wireless security simulation.

Theory - A wireless local Area Network allows devices to connect to a network using radio signals instead of physical cables. Because wireless signals travel through open air, they are more vulnerable to unauthorized access and attacks. Therefore, strong security mechanisms are required to protect the network.

key wireless security mechanisms used.

1. WPA 2 / WPA 3 security

- WPA 2 uses AES encryption, which is strong and widely used.
- WPA 3 is the latest and more secure standard providing:
- Better protection against password-guessing attacks.
- Strong encryption
- Improved security for open networks.

2] MAC address filtering

- Every device has a unique MAC address.
- The access point is configured to allow only approved MAC addresses.

- This adds an extra layer of security by restricting unauthorized devices.

3) SSID Management

- Service set identifier is the wireless network name.
- Security practice includes:
 - Hiding the SSID
 - Using a unique, non-predictable SSID name
 - Avoiding default SSID provided by router.

4) Wireless attacks simulated common attacks.

- Eavesdropping
- Rogue access point deauthentication attack.

5) Countermeasure configured

to protect the WLAN

- Stronger WPA 2 / WPA 3 encryption.
- Complex wifi password
- MAC filtering enabled
- SSID hidden

Wireless-N Broadband Router

Firmware Version: v0.93.3

Wireless-N Broadband Router WRT300N

- Wireless**
- Setup**
- Security**
- Wireless Security**
- Wireless Settings**
- Basic Wireless Settings**
- Access Restrictions**
- Guest Network**
- Wireless MAC Filter**
- Applications & Gaming**
- Advanced Wireless Settings**
- Administration**
- Status**

Wireless Security

WPA2 Personal

AES

StrongPassword123

3600

Help...

6. Impact on connectivity and performance.

- Higher security (WPA 3 + MAC filtering + hidden SSID) → Better security but slightly slower speed or connection time.

- Lower security → Faster connection but highly insecure.

Conclusion: We successfully deployed and secured a wireless LAN using advanced security mechanisms such as WPA 21, WPA 3, MAC filtering, SSID management. We simulated common wireless attacks and implemented effective countermeasures demonstrating that stronger security improved network protection but may slightly impact connectivity & performance.

~~24/2/26~~
24/2/26

Aim : Multilayer switch configuration.

problem statement : Design an IPv6-based network addressing, plan and configure IPv6 addressing on routers and hosts. Ensure proper communication using link-local, global unicast, and auto-configuration methods (SLAAC). Test IPv6 connectivity using ICMPv6 and neighbor discovery tools.

Theory : Multilayer switch configuration

A multilayer switch (MLS) is a network device that performs both:

- Layer 2 (Data link layer) → switching using MAC addresses.
- Layer 3 (Network layer) → Routing using IP addresses.

It combines the functionality of a switch and a router in one device.

This improves speed, efficiency, and performance in modern networks.

21 objective of the experiment

- configure VLAN's
- configure SVI's (Switch virtual Interfaces)
- enable Inter-VLAN routing
- configure routing protocol
- Implement Port security
- verify switching and routing operation.

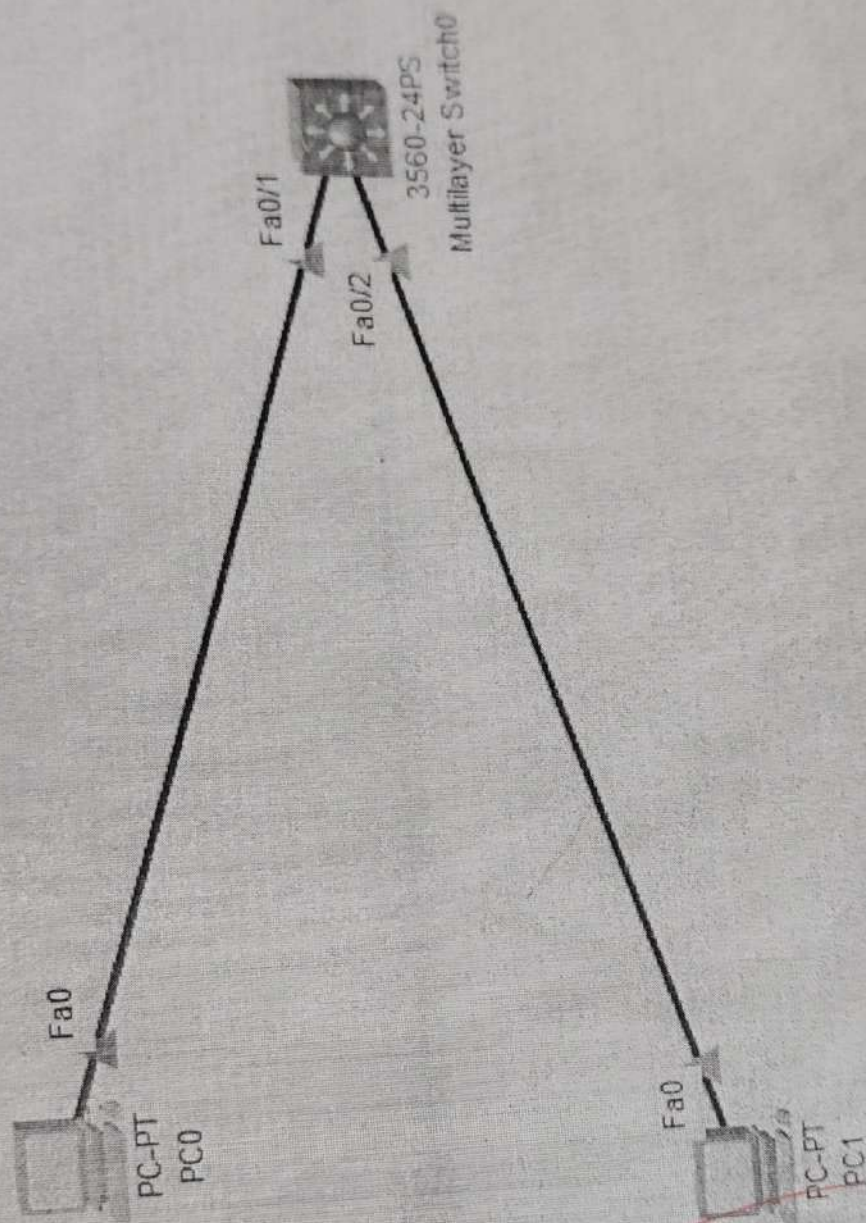
Layer 2 switching - MAC Address, Frame forwarding, VLAN Segmentation.

Switch stores MAC addresses in MAC address table.

- Functions :
- Reduces collision domain
 - Improves network performance
 - segments network using VLAN's

VLAN (virtual local area network)

- A VLAN logically divides a physical network into multiple separate broadcast domains.
- Improves security.
 - Reduces broadcast traffic.
 - logical grouping of devices.



```
Switch>
Switch>enable
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.2.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

Class Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1
Invalid Command.

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.1:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=3ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>

```

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTRL/Z.
Switch(config)#ip routing
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#interface vlan 20
Switch(config-if)#
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.2.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#

```

Copy

Paste

```
Switch>enable
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.2.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

```

Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.2.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#ipp dhcp excluded-address 192.168.10.1
Switch(config)#no shutdown

```

```

% Invalid input detected at '^' marker.

```

```

Switch(config)#ip dhcp excluded-address 192.168.10.1
Switch(config)#ip dhcp excluded-address 192.168.20.1
Switch(config)#ip dhcp pool vlan10
Switch(dhcp-config)#network 192.168.10.0 255.255.255.0
Switch(dhcp-config)#default-router 192.168.10.1
Switch(dhcp-config)#exit
Switch(dhcp-config)#exit
Switch(config)#dhcp pool vlan20

```

```

% Invalid input detected at '^' marker.

```

```

Switch(config)#ip dhcp pool vlan20
Switch(dhcp-config)#network 192.168.20.0 255.255.255.0
Switch(dhcp-config)#default-router 192.168.20.1
Switch(dhcp-config)#exit
Switch(config)#

```

Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=2ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=3ms TTL=255

Ping statistics for 192.168.20.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address: FE80::20A:41FF:FEE4:47C2
IPv6 Address: ::
IPv4 Address: 192.168.10.10
Subnet Mask: 255.255.255.0
Default Gateway: ::
192.168.10.1

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address: ::
IPv6 Address: ::
IPv4 Address: 0.0.0.0
Subnet Mask: 0.0.0.0
Default Gateway: ::
0.0.0.0

Bluetooth Connection:

```
Connection-specific DNS Suffix...:
Link-local IPv6 Address...:::
IPv6 Address...:::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...:::
0.0.0.0
```

C:\>ping 192.168.30.1

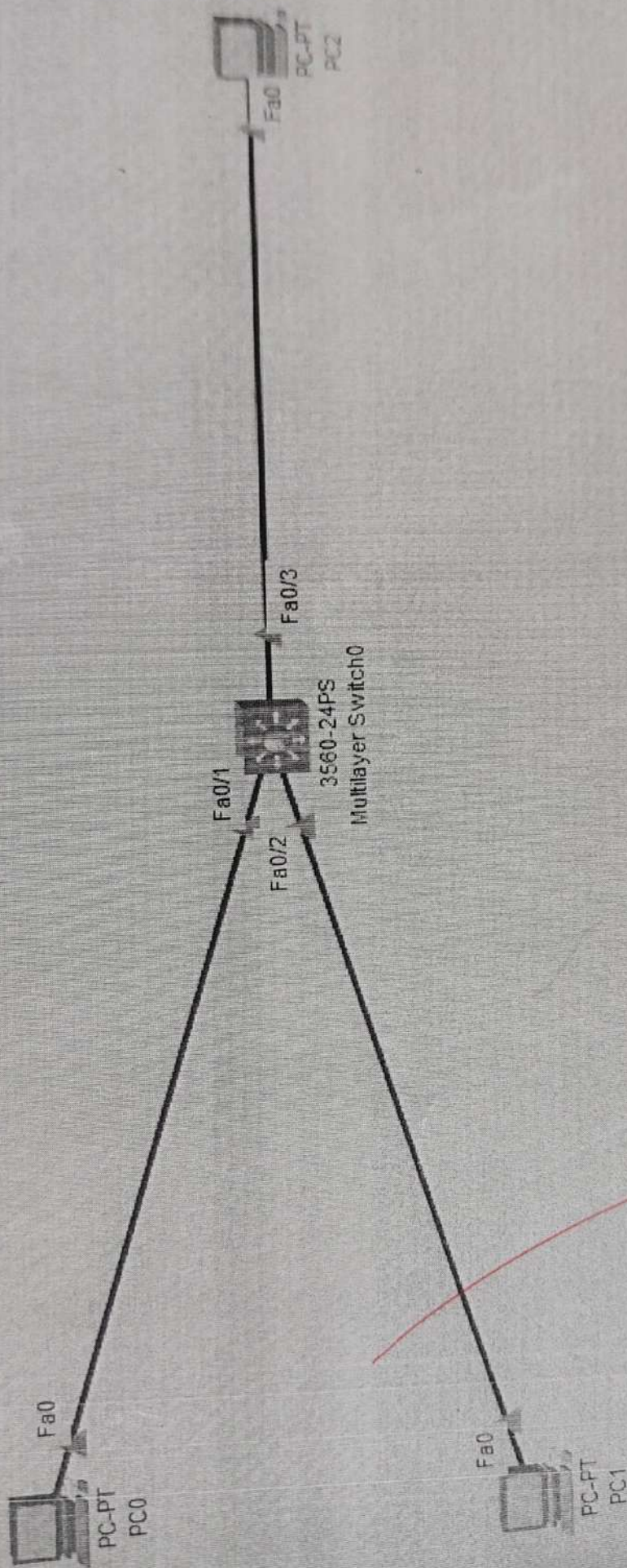
Pinging 192.168.30.1 with 32 bytes of data:

```
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
```

Ping statistics for 192.168.30.1:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>



- ex: • VLAN 10 → sales department
• VLAN 20 → HR department

Devices in different VLAN's cannot communicate directly without routing.

SVI (switch virtual interface)

SVI is a virtual layer 3 interface created for a VLAN.

- assigning IP address to VLAN
- performing inter-VLAN routing.

ex: interface VLAN 10

IP Address 192.168.10.1 255.255.255.0

high-performance inter-VLAN communication.

Multilayer switch performance routing in hardware using ASIC.

- ultra-speed routing
- Reduced CPU load
- Faster packet processing.

Conclusion: Hence, we studied Multilayer switch configuration.

Aim: IPv6 network setup

problem statement: Design an IPv6-based network addressing, Plan and configure IPv6 addressing on routers and hosts. Ensure proper communication using link-local, global unicast, and auto-configuration methods (SLAAC). Test IPv6 connectivity using ICMPv6 and neighbor discovery tools.

Theory: IPv6 (Internet Protocol version 6) is the latest version of IP address system used to identify devices on a network.

• IPv4 example: 192.168.1.1

• IPv6 example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv6 is created because IPv4 addresses were getting finished.

why do we need IPv6?

- Huge number of addresses
- Better security
- Faster routing
- Built-in auto configuration.

Types of IPv6 Addresses

(1) Link-Local addresses

- Start with $FE80::$
- Used for communication inside same network.
- Automatically created
- Cannot be used on internet.

Ex: $FE80::1$

(2) Global Unicast Address

- Used for internet communication
- Public addresses
- Similar to public IPv4.

Ex: $2001:DB8::1$

(3) Unique Local Address

- Similar to private IPv4
- Starts with $FC00::$ or $FD00::$

IPv6 network Addressing plan (Design)

- 1 Router
- 2 LAN network
- 4 PCs



user>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip v6 unicast-routing

Router(config)#interface g0/0

Router(config-if)#ip v6 address 2001:db8:1:1::1/64

Router(config-if)#no shutdown

Router(config-if)#

Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

EXIT

Router(config)#interface g0/1

Router(config-if)#ip v6 address 2001:db8:1:2::1/64

Router(config-if)#no shutdown

Router(config-if)#

Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

EXIT

Router(config)#

nsco Packet Tracer PC Command Line 1.0

>PING 2001:DB8:1:2::1

Pinging 2001:DB8:1:2::1 with 32 bytes of data:

Reply from 2001:DB8:1:2::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1:2::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1:2::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1:2::1: bytes=32 time<1ms TTL=255

ping statistics for 2001:DB8:1:2::1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

:A>

LAN 1 $\rightarrow$ 2001:DB8:1::/64 $\rightarrow$ 2001:DB8:1::1
LAN 2 $\rightarrow$ 2001:DB8:2::/64 $\rightarrow$ 2001:DB8:2::1

Configure IPv6 on PC

Manual Configuration - Enter IPv6 address

- enter prefix /64
- enter default gateway.

Method 2: SLAAC (Auto Configuration)

SLAAC = Stateless Address Auto Configuration

Router automatically sends address info

PC automatically generates its own IPv6 address

No DHCP server needed

Router sends Router Advertisement (RA) message.

PC combines:

• Network prefix (from router)

• Interface ID (from MAC address)

then creates full IPv6 address automatically.

Conclusion: Hence, we studied IPv6 network setup, is the latest internet protocol that overcomes the limitations of IPv4 by providing a very large address space.

Aim: IPv6 Routing Configuration.

problem statement: Implement dynamic IPv6 routing protocols such as RIPv6, OSPFv3, or EIGRP for IPv6 in a multi-router environment. Ensure that all routers interconnect successfully and achieve full network reachability. Validate route propagation and forwarding behavior.

Theory: IPv6 routing means sending data (packets) from one network to another using IPv6 addresses.

- Just like google maps shows routes between places, routers find the best path for data.
- Routers use routing protocols to share information with each other.

Why do we need dynamic routing?
There are two types of routing:

1. Static Routing

- you manually enter routes.
- works for small networks.

2. Dynamic Routing

- routers automatically learn routes.
- if network changes $\rightarrow$ routers adjust automatically.
- saves time and reduces errors.

Main IPv4 Routing protocols.

1. RIPv2 (Routing Information Protocol next generation).

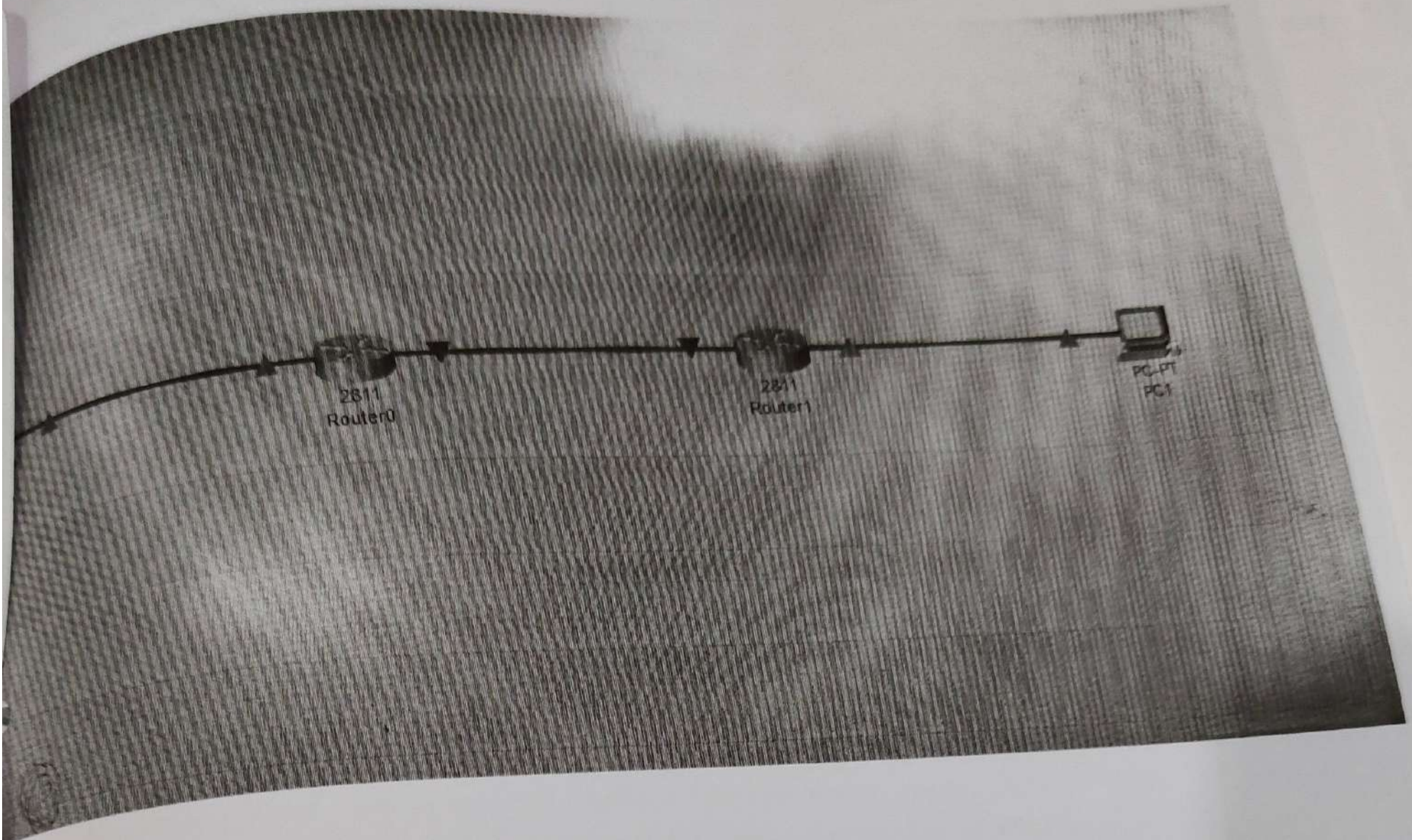
features: uses hop count (number of routers) to find best path.

ARISE & SHINE

- maximum 15 hops allowed
- slow convergence (takes time to update changes)

best for: small networks.

ex: If Router A $\rightarrow$ Router B $\rightarrow$ Router C
Hop count = 2 $\rightarrow$ shortest path chosen.



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z
Router(config)#ipv6 unicast-routing
Router(config)#interface fastethernet0/0
Router(config-if)#ipv6 address 2001:DB8:1::1/64
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config)#interface fastethernet0/1
Router(config-if)#ipv6 address 2001:DB8:2::1/64
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

```
Router>
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface fa0/0
Router(config-if)#ipv6 address 2001:DB8:12::2/64
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
```

```
Router(config)#interface fa0/1
Router(config-if)#ipv6 address 2001:DB8:2::1/64
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
```

```
Router(config)#ipv6 route 2001:DB8:1::/64 2001:DB8:12::1
Router(config)#
```

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```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface fa0/0
Router(config-if)#ipv6 address 2001:1::1/64
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

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of OSPF v3 (open shortest path first version 3)
features: • uses cost (bandwidth) instead of hop count.

- Faster than RIPv2
- supports large networks
- uses link-state algorithm.

Adv: • very fast convergence.
• efficient routing.

Best for: Medium to large networks.

Validation of Routing.

To check if everything is working:

use commands:

- ping → check connectivity.
- tracroute → check path
- show ipv6 route → check routing table.

Conclusion: Hence, all directed IPv6 Routing Configuration.

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Aim : A-A-A Configuration.

Problem Statement: Configure authentication, Authorization, and Accounting (AAA) services using a TACACS + or RADIUS server. Implement centralized login control for network devices. Test user authentication, command authorization, and activity logging to ensure secure and control administrative access.

Theory : AAA stands for:-

A → Authentication

A → Authorization

A → Accounting

It is used to control and secure access to network devices like routers and switches.

1. Authentication (Who are you?)

This step checks user identity.

User enters username + password

System verifies if the user is valid or not. ex: like logging into your phone using password or fingerprint.

2. Authorization (What can you do?)

After login, it decides what permissions user has:-

1. which command user can run.

2. what access level they get.

ex:- Admin → full control

• user → limited access.

3. Accounting (What did you do?)

This step keeps record of user activity.

• when user logged in/out

• what command they executed

• how long they stayed.

ex: like history / log of all actions.

What is centralized AAA?

Instead of storing users on each router, we use a central server.

Two main server:-

• TACACS + (Terminal Access Controller Access - control system plus)

• RADIUS (Remote Authentication Dial - in user service)

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.0

Pinging 10.0.0.0 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.

Request timed out.

Reply from 192.168.1.1: Destination host unreachable.

Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 10.0.0.0:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

I

```

Router(config)#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Router(config)#aaa ?
accounting      Accounting configurations parameters.
authentication  Authentication configurations parameters.
authorization    Authorization configurations parameters.
new-model       Enable NEW access control commands and functions. (Disables
                OLD commands.)

```

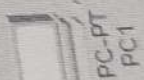
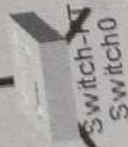
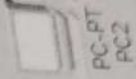
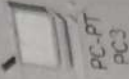
```

Router(config)#aaa new-model
Router(config)#username admin
Router(config)#aaa authentication ?
enable          Set authentication lists for enable.
login           Set authentication lists for logins.
ppp             Set authentication lists for ppp.
Router(config)#aaa authentication login
? Incomplete command.
Router(config)#aaa authentication login default ?
enable          Use enable password for authentication.
group           Use Server-group.
local           Use local username authentication.
local-case     Use case-sensitive local username authentication.
none           NO authentication.
Router(config)#aaa authentication login default ?
enable          Use enable password for authentication.
group           Use Server-group.
local           Use local username authentication.
local-case     Use case-sensitive local username authentication.
none           NO authentication.
Router(config)#aaa authentication login default local
Router(config)#aaa authentication login default local
Router(config)#enable password cisco
Router(config)#line console 0
Router(config-line)#login
* You can only use the command "[no] login authentication ..." when aaa is enabled.
Router(config-line)#login authentication default
Router(config-line)#pass

```

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Destination

Source

Last Status

File

Scenario 0

[illegible]

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purpose of AAA.

- improves security
- provides central control
- tracks user activities.
- prevents unauthorized access.

How AAA works.

1. user tries to login.
2. Device sends request to AAA server
3. server checks username & password.
4. if correct → access granted
5. server decides permission
6. All activities are recorded.

AAA is a security framework used to control access to network devices by verifying user identity (Authentication), giving permissions (Authorization), and recording activities (Accounting).

Conclusion : Hence, we studied AAA configuration.

~~##~~
20/11/20

Aim: Bridge connection between Routing protocols.

problem statement: Integrate two or more different routing protocols in a single network and configure route redistribution between them. Ensure seamless communication across domains running protocols such as RIP, EIGRP, and OSPF. Validate proper redistribution while avoiding routing loops and suboptimal paths.

Theory: Bridge connection between routing protocols: Routing protocol is a set of rules that routers use to decide the best path to send data from one network to another.

- RIP (Routing Information protocol)
- EIGRP (Enhanced Interior Gateway Routing protocol)
- OSPF (Open Shortest Path first)

Sometimes, different parts of a network use different routing protocols.

Ex: • one network uses RIP
• another network uses OSPF
These networks cannot understand each other directly.

Solution: we use something called route redistribution.
It acts like a bridge (connection) between different routing protocols.

What is Route Redistribution?

Route Redistribution means sharing routes from one routing protocol to another.

Ex: • Routes from RIP are shared with OSPF.
• Routes from OSPF are shared with EIGRP.
So, all networks can communicate easily.

Why is it needed? → To connect different networks using different protocols.

- To allow smooth communication (seamless communication)
- To make a large network work as one system.

How it works:-

- (1) A router runs multiple routing protocols.
 - (2) It takes routes from one protocol.
 - (3) Converts them into another protocol format.
 - (4) Shares them with the other network.
- This router is called a Redistribution Router.

Real-life example $\rightarrow$ RIP, OSPF

Route Redistribution = Translator.

It helps both languages understand each other and communicate.

How to avoid problems?

- Use proper metrics (cost values)
- Use route filtering (control which routes are shared)
- Avoid redistributing the same route again and again.

Conclusion: Hence, we studied bridge connection between routing protocols.

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=12ms TTL=126

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Router>enable
Router#configure terminal

Router(config)#interface g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%IP-4-DUPADDR: Duplicate address 192.168.2.1 on GigabitEthernet0/0, sourced by
0005.5E63.652B

Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```
Router ospf 1
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#router ospf 1
Router(config-router)#redistribute rip subnets
Router(config-router)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#redistribute ospf 1 metric 2
Router(config-router)#exit
Router(config)#exit
```

```
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
```

```
Router(config-if)#
```

```
LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
```

```
Router ospf 1
```

```
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
```

```
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
```

```
Router(config-router)#router ospf 1
```

```
Router(config-router)#redistribute rip subnets
```

```
Router(config-router)#router rip
```

```
Router(config-router)#version 2
```

```
Router(config-router)#no auto-summary
```

```
Router(config-router)#network 10.0.0.0
```

PC-PT
PC0

2911
Router0

2911
Router1

PC-PT
PC1